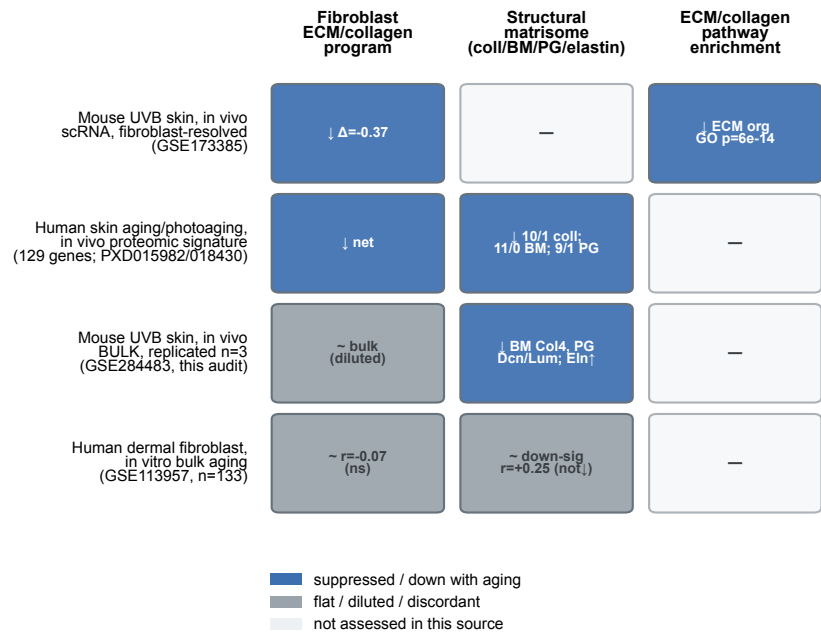


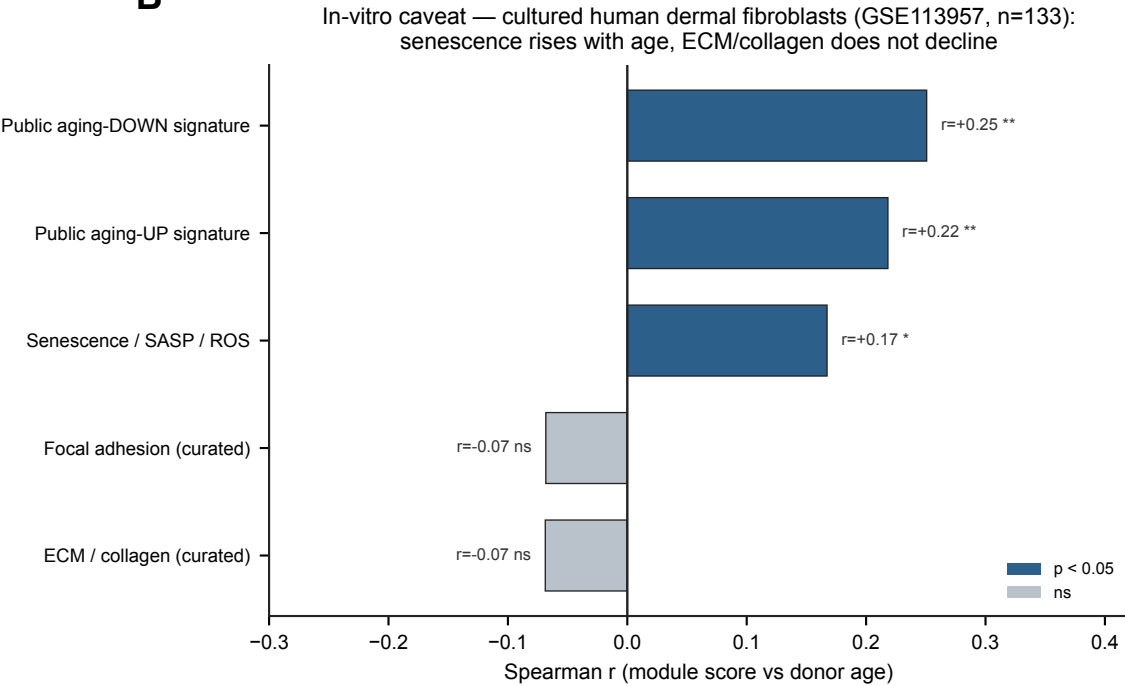
Supplementary public-data validation: cross-source convergence on ECM/collagen decline, replicate-aware and honestly bounded

Independent public evidence converges on ECM/collagen decline in vivo — honestly bounded in vitro and in bulk

A

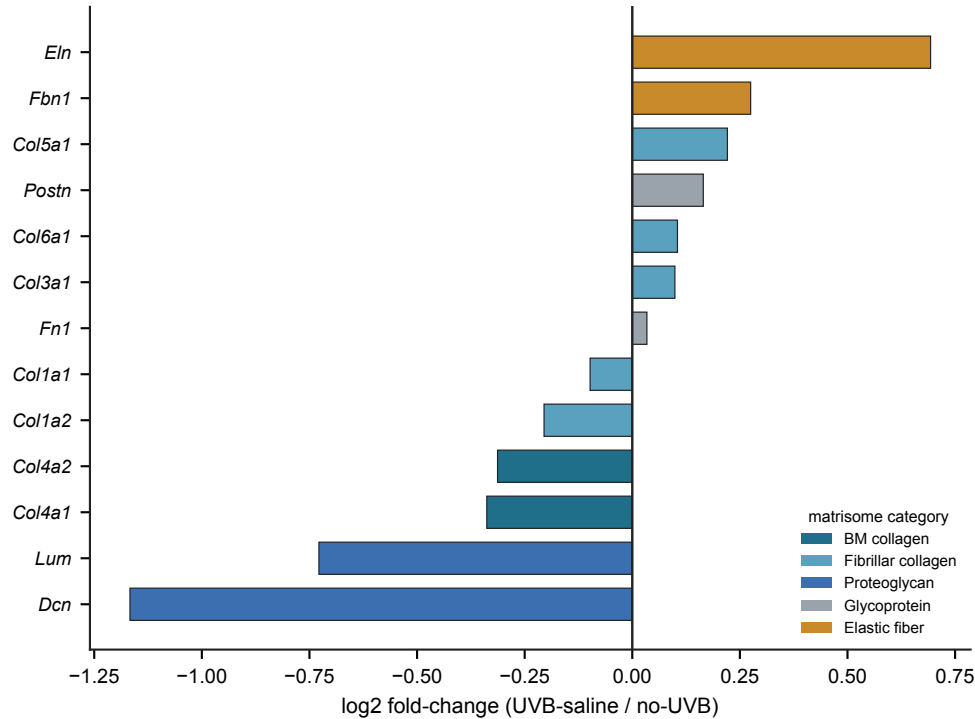


B



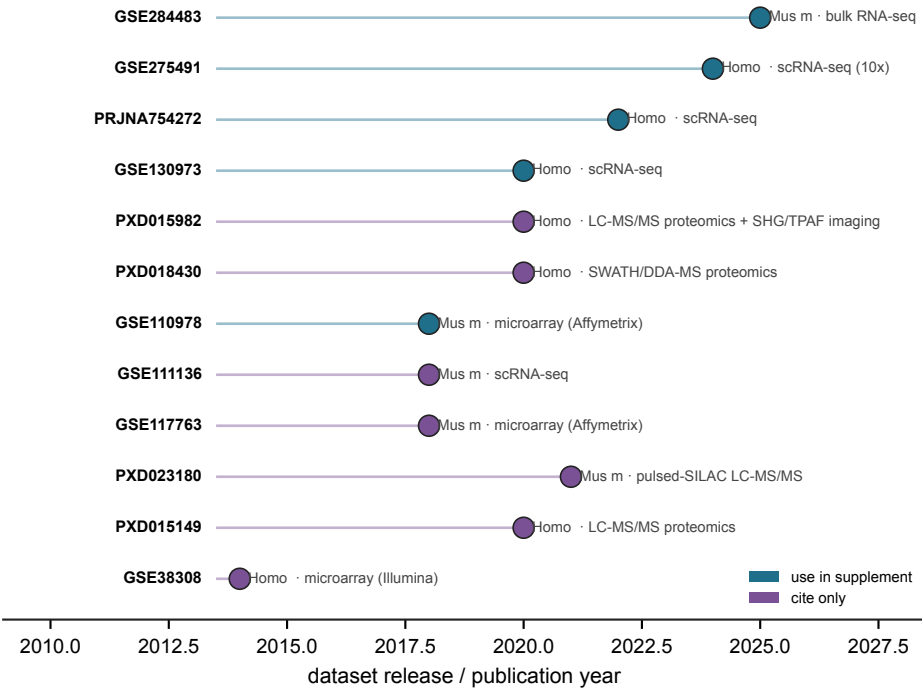
C

NEW replicate-aware in-vivo reanalysis — mouse UVB skin (GSE284483, n=3/group): basement-membrane collagens + proteoglycans down, elastin up (solar elastosis)



D

Screened public datasets (top-ranked) and recommended use



Public-data only (no experimental data from this study). GSE113957: replicate-aware (Spearman vs donor age, n=133). GSE284483: replicate-aware (n=3 vs 3, Welch t). GSE173385 module difference is exploratory (cell-level; one biological sample per condition).

whole-skin module-level n.s. (Welch t $p=0.93$, n=3 vs 3): the fibroblast-specific suppression is diluted in bulk tissue — consistent with the Panel A bound.

Full screening of 33 datasets in public_dataset_candidates_2022_2026.tsv